Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Certifiably Safe and Precise Control of AI-Infused Robotic Systems with Vision and Language: Theory and Experiments			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (✓), Hydrogen ( ), Cybersecurity (✓), AI (✓), Next-gen Communication ( ), Advanced Robotics & Manufacturing (✓), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Control and Intelligence Technologies	Aerospace Systems	Aerospace System Software
Concept of Proposed Technology	Recent advances in AI and foundation models (FMs) have shown potential for end-to-end design and operation of fully autonomous robotic systems. However, controlling their intent and long-term impact safely with high precision in high-consequence scenarios still relies on domain-specific heuristics. Control theory attempts to enforce performance-critical system behavior through numerical data, yet the information processed in AI and FMs is not just numbers. As in human cognition, their billion-parameter scale, dynamic, and nonlinear intersystem interactions abstract real-world context into semantics, rendering the models black boxes for numerical and model-based analyses. We are developing a novel control theory that will natively integrate visual and language signals into arbitrary AI- and FM-infused robotic systems. Crucially, even with these signals in the loop and with limited knowledge of real-world hazards, our theory will offer rigorous closed-loop guarantees—robots can be controlled to converge **exponentially fast with pre-certifiable precision** to arbitrary desired states, where the states can also be specified via vision and language. A partial list of our robots can be found here; they can be deployed for rapid-cycle theory development and validation in our large-scale experimental facility, which features a 120 ft x 60 ft x 20 ft motion-captured space.			

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Rationale for Joint R&D Project	This work is part of our ongoing effort to establish a world-leading research initiative and a flagship testbed for end-to-end autonomy in aerospace robotics within our institute. In collaboration with Korean SMEs, this effort can be substantially strengthened by incorporating state-of-the-practice perspectives on engineering pathways, productization expertise, and application-driven requirements that are rarely accessible in academia. Our group will provide foundational research leadership and rapid-cycle experimental infrastructure, delivering state-of-the-art capabilities to establish certifiable autonomy across arbitrary robotic systems. Joint execution will enable rapid iteration between theory, AI and FM integration, and hardware demonstrations under industrially realistic constraints, accelerating translation into competitive autonomy and robot control technologies with provably measurable safety and precision. We anticipate that our end-to-end research, validation, and technology development pipeline, together with the visually demonstrated capabilities arising from this joint project, will make our contributions readily accessible to researchers worldwide beyond academia, including those outside our field. Supported by the Center for Autonomy, this effort will enable academic collaborators, government stakeholders, investors, and industrial partners to map their own problem formulations and application needs onto our demonstrated capabilities, catalyzing new interdisciplinary projects and sustained innovation.
Keywords	1. Certifiable Autonomy in AI- and Foundation Model-Infused Systems 2. High-Precision Robot Control with Semantic Hazard Avoidance 3. Statistical Machine Learning and Nonlinear Control with Non-Traditional Data 4. Safe and Secure Decision-Making in High-Consequence Aerospace Systems See here for more information.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) 2)	**Start of the Project**	Level 2	**End of the Project**	Level 4 – 5

R&D Objectives

Our group's technology centers around the statistical and structural abstraction of *contraction theory* [1] – [3], which provides a dynamically intrinsic concept for formally characterizing how control systems exponentially align with their ideals. In essence, given completely black-box systems—spanning robots, autonomous vehicles, communication networks, financial trading services, and more—together with their observable data in their real-world operations, our framework will characterize and enhance the theoretical limits of performance guarantees achievable through control. The guarantees here can be calibrated to match arbitrary users' requirements on trustworthiness and fidelity (e.g., system safety, information security, and control accuracy). We have tested our framework on some of the robotic platforms listed in Fig. 1 [4] – [6], in distributed multi-agent spacecraft coordination [7], and in interstellar exploration [8] in collaboration with the NASA Jet Propulsion Laboratory.

Fig. 1: Available robotic resources, where the background image shows the Stock Pavilion with a 120 ft × 60 ft ×20 ft motion-captured space. (Image credit: Flapper Drones, bionicbird, ModalAI, Hiwonder, Holybro, DJI, Wheeltec, Robolink, Bitcraze, Dell, Nvidia, Delta AI, OptiTrack, Unitree Robotics, and Agilex Robotics. See https://hirotsukamoto.com/robots/.)

Quantitatively, our recent work [2], [6] applies these ideas to enforce robustness, stability, safety, and incremental stability with high precision across a wide range of black-box nonlinear dynamical systems, as shown in Figs. 2 – 4. Importantly, these figures indicate that **our approach enables all performance curves to converge exponentially to their ideal scenarios using only observable system inputs and outputs, even with no structural or distributional knowledge of the underlying systems**. An open-source implementation with a tutorial is partially available here: https://hirotsukamoto.com/contraction/.

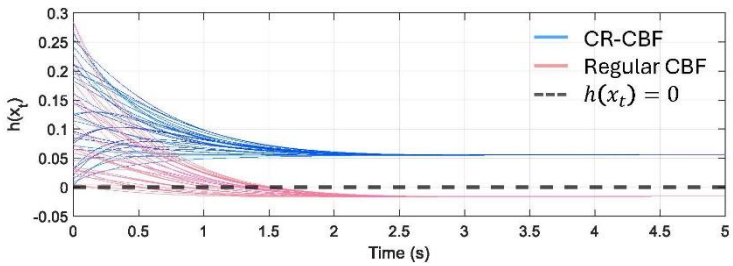


Fig. 2: CR-CBF (our approach) applied to autonomous adaptive cruise control, ensuring safety by maintaining operation above the dashed line [2].

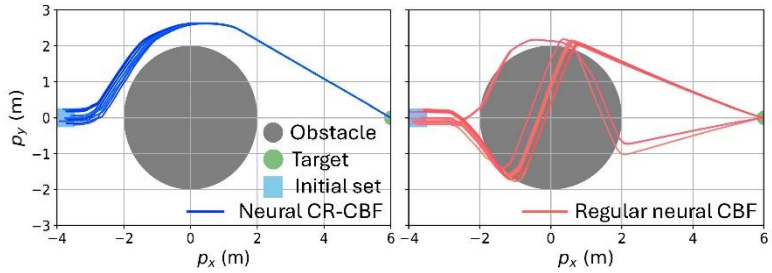


Fig. 3: Neural CR-CBF (our approach) applied to Dubins car collision avoidance [2].

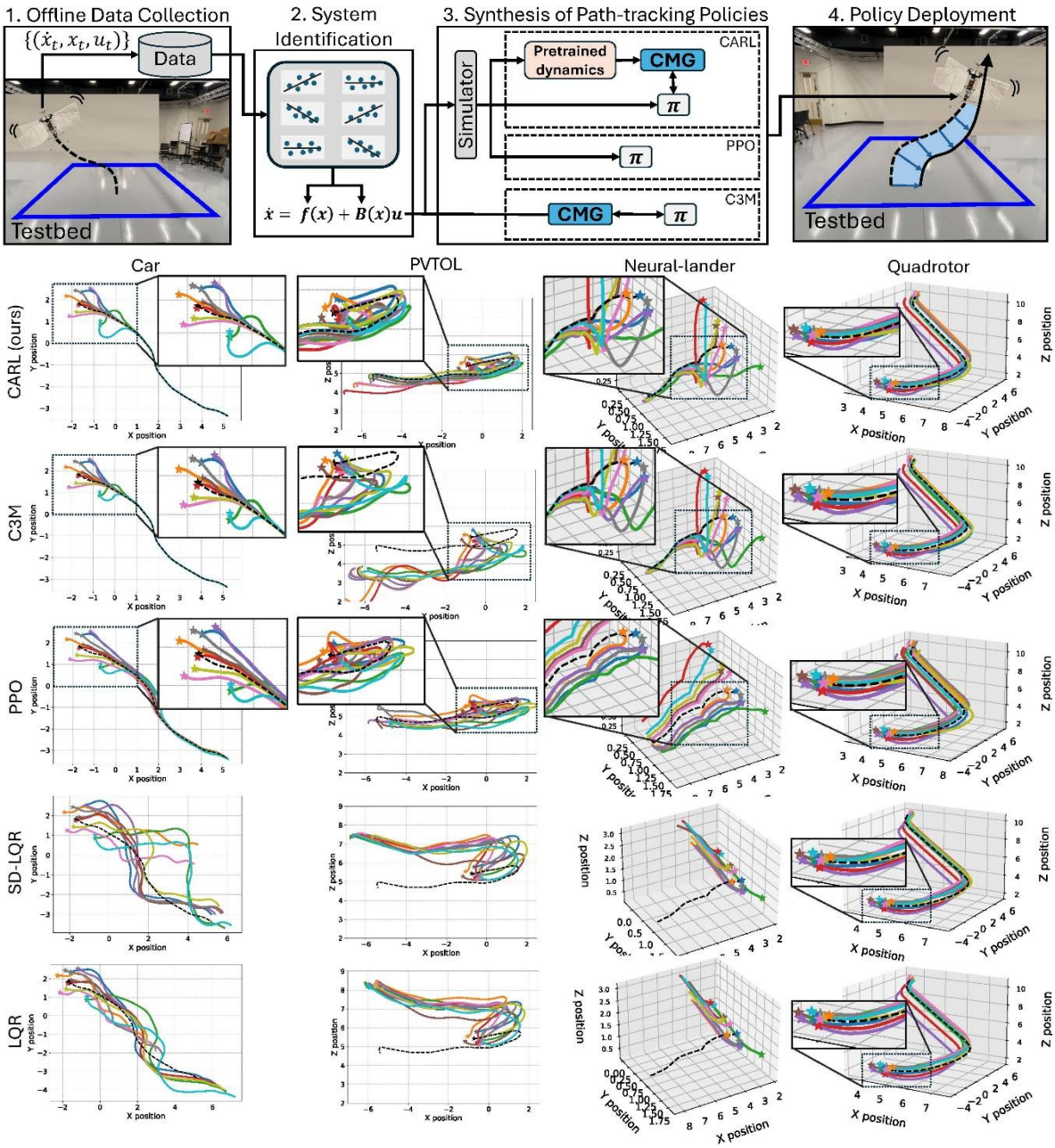
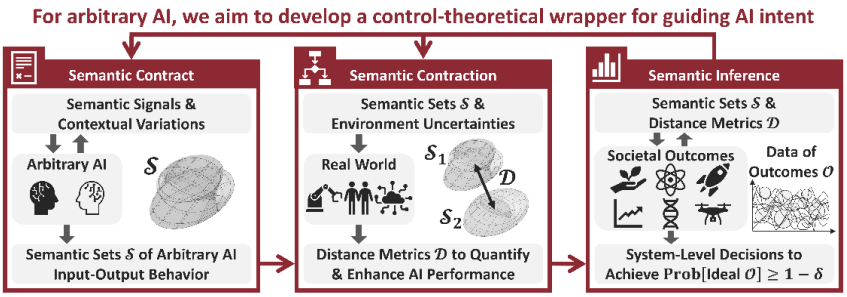
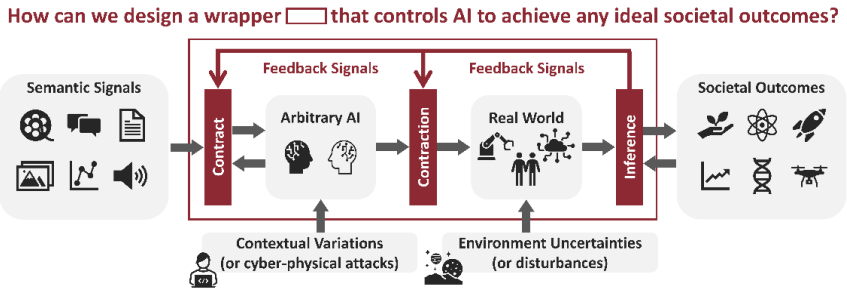


Fig. 4: Our approach with reinforcement learning (CARL) applied to a variety of black-box nonlinear dynamical systems [6].

Naturally, in our ongoing projects, we are developing high-performance control technologies for arbitrary black-box dynamical systems, now infused with AI and foundation models (FMs). This is inspired by human cognition, where our cognition abstracts real-world context non-numerically, linking observations not merely to data patterns but to intent and semantics. **We have preliminary results that accommodate abstract vision and language signals in the design of our high-precision control for completely black-box systems.** Therefore, the primary objective of this project is a scaled-up demonstration of this capability in black-box

	systems with vision- and language-infused AI and FMs. Using the Center for Autonomy (CfA) facilities and our GPU data center, we believe we can reproduce a variety of high-consequence robot operation settings, including (i) vision- language-based precision autonomous landing under degraded visibility on cluttered terrain, (ii) human–heterogeneous robot collaboration relying solely on verbal communication, and (iii) time-critical environment monitoring and mapping with semantic hazard avoidance, all of which rely heavily on semantic signals. The involvement of Korean SMEs is nearly essential in this effort. While our group has strong expertise in the theory and design of certifiable autonomy with vision and language, we do not claim comparable depth in identifying, scoping, and prioritizing the highest-impact application pathways through which these technologies can most directly benefit society. Korean SMEs, with their domain knowledge of operational constraints and proximity to real-world deployment needs, can play an instrumental role in defining application scenarios of our technology that properly reflect industrial and societal value. Accordingly, achieving the objective stated earlier will depend critically on establishing SME-driven use cases and evaluation benchmarks. The prevailing pursuit of empirical success in AI- and FM-infused robots, without adequate theoretical justification, is especially risky for real-world autonomy—e.g., FM-infused aircraft that misclassify a runway in low visibility, or crewed spacecraft with AI agents that misinterpret docking cues—catastrophically entangling robot decisions with spurious intent and hallucinations in AI and FMs [9], [10]. This project aims to fundamentally address this issue by establishing a certifiable theoretical foundation and real-world tested capabilities for AI- and FM-infused autonomy.
R&D Target Specifications	Project success will be assessed through three major milestones specified as follows: 1. Rigorous theory development and large-scale experiments, culminating in multiple top-tier journal and conference publications (at least one strong publication per year) for the broad dissemination of our results. 2. Selection of two to three high industrial and societal-impact application scenarios that leverage our theory with vision–language AI and FMs, aligned with the priorities of the participating SMEs. 3. Full-scale validation of the theory in the selected scenarios, demonstrating order-of-magnitude gains in control precision and in safety guarantees, targeting TRL 4 – 5.
R&D Plan	The theoretical component of this effort builds on our ongoing development of *contextual control theory*. The theory will comprise three layers: *semantic contract*, *contraction*, and *inference* to (i) symbolically prescribe how vision- and language-infused systems will behave across uncertainties, (ii) derive system states to exponentially converge toward vision- and language-specified desired states, and (iii) leverage system input–output data with (i) to enable certifiably safe and high-precision control in (ii). Figs. 5 and 6 visualize these core technical concepts as a wrapper for controlling the intent of arbitrary AI systems. Again, this theoretical development will be followed by empirical validation in high-impact societal applications aligned with the participating SMEs' priorities, which constitutes the primary objective of this project.  *Fig. 5: Conceptual overview of contextual control theory.*  *Fig. 6: Technical overview of contextual control theory.*

Thrust 1—Semantic Contract: We will design transformer-based distance metrics to quantify similarities and discrepancies among semantic signals in the inputs and outputs of arbitrary AI and FMs. This may be achieved by fine-tuning existing signal-type-specific transformers on Hugging Face (e.g., language [11], vision [12], or audio [13]). Using such metrics, we will generate datasets of signal-type-specific contextual variations, i.e., sets of distinct contexts leading to semantically equivalent signals (e.g., variations in illumination for visual data, or paraphrased expressions for linguistic data). They could also include context-modifying cyber-physical perturbations, as in Fig. 6.

This process is conceptually analogous to determining the "Lipschitz constant" of AI and FMs in our transformer metrics against contextual variations. Our objective is to construct a streamlined framework here, which will serve as a contract between systems and users: when integrated with Thrust 3, this will provide a statistical guarantee that the AI and FMs will achieve a specified level of performance with respect to our transformer-based metric. This may be viewed as a generalization of [14] to semantic signals and will be referred to as a *semantic contract*.

Thrust 2—Semantic Contraction: Given the semantic contract, we can characterize compact sets of semantic signals within which the outputs of the AI and FMs reside, with the size of the sets determined by the transformer-based metrics. Using these semantic sets to define the states of AI and FMs, we will train another transformer for predicting their evolutions (e.g., as in [15]), assuming access to or the ability to generate a dataset of AI- and FM-infused dynamics in the real world. The control inputs here will be adjustable interventions to the inputs or outputs of the AI and FMs. Inspired by contraction theory [1], we hypothesize that such a dynamical model may enable the systematic design of a contraction metric and a contracting control policy. Such metrics and controllers can be synthesized through convex optimization or inferred directly from data, exhibiting robustness to structured environment uncertainties and disturbances. Our objective is to verify this hypothesis. This semantically data-driven controller design will be referred to as *semantic contraction*.

Thrust 3—Semantic Inference: What is missing in Thrusts 1 and 2 is the rigorous validation of the constructed metrics, without which the entire contract and contraction guarantees would remain merely empirical. We will thus construct a statistical inference module to measure contraction violation, encompassing modeling uncertainties in the previous thrusts, as well as all the other unstructured uncertainties that may be present in real-world environments. This will be evaluated using actual semantic sets observed through real-world interactions. If such a score can be successfully defined, our work on statistical inference [2] – [5] will establish formal probabilistic guarantees. In particular, with meta-adaptive control [16], the guarantees will enable continual improvement and trustworthy redesign of AI and FMs within the wrapper. This process will be referred to as *semantic inference*.

Together, these thrusts will establish a pre-certifiable and implementable pathway for driving AI and FM-infused systems toward their semantic sets of ideals, which can be calibrated to SMEs' critical operational requirements or broader needs for societal-scale innovations.

Thrust 4—Full-Scale Validation: We will empirically validate our framework by replacing the arbitrary AI and FMs in Fig. 6 with those from Hugging Face. As noted earlier, a central part of this effort will be the selection of two to three high-impact industrial and societal application scenarios, aligned with the priorities and expertise of the participating SMEs. Within the selected scenarios, we will demonstrate order-of-magnitude improvements in control precision and in safety guarantees, targeting TRL 4 – 5. As one example—purely as a possible direction—we could integrate the Prithvi Earth FMs [15] to simulate the dynamics of disaster evolution and climate change, with Hugging Face AI/FMs serving as decision-making modules. By defining the semantic ideal as mitigated disaster or stabilized climate states, this platform would serve as a testbed for demonstrating the practical applicability of our framework to societal-scale, AI- and FM-infused dynamical systems. We can also combine these ideas with the high-consequence robotic operation settings described earlier.

Risks and Mitigation: The proposed effort will demand rigorous theoretical development and substantial computational resources. To expedite progress, Thrust 1 will begin with well-supported data types from prior work (e.g., language [11], vision [12], or audio [13]) and numerically embedded contracts [14]. Such embeddings should work nominally with our prior results on statistical contraction [2] – [5], supporting initial development in Thrusts 2 and 3. For computation, **our group has exclusive access to a Dell PowerEdge R7625 server with NVIDIA L40S and on-demand access to the Delta AI campus cluster with NVIDIA H100 for rapid testing and training**, before scaling up with commercial tools like SageMaker and EC2. These will facilitate quick

validation prior to full-scale deployment despite the ambitious scope of this project.

As shown in Table 1, the seed round (Phase 1) will be used primarily for nominal theory development and for selecting target application scenarios for our joint proposal. Using our Center for Autonomy facilities, we can also perform an initial proof-of-concept validation, particularly if the selected application is robotics-oriented. Table 2 shows the timeline for the full funding round (phase 2).

Table 1: Project timeline, where M *denotes months.*

	M1	**M2**	**M3**	**M4**	**M5**	**M6**
Application Selection	Led by SMCs					
Nominal Theory Development	Led by UIUC					
Initial Concept Validation				Jointly by UIUC and SMCs		

Table 2: Project timeline, where Q *denotes quarters.*

	Q1	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
Thrust 1	Led by UIUC									
Thrust 2			Led by UIUC							
Thrust 3					Led by UIUC					
Thrust 4		Jointly by UIUC and SMCs								

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	As described above, our group will lead the theoretical development and the large-scale concept demonstration using the Center for Autonomy facilities and the GPU and CPU data center. If the selected applications involve robots, we can also provide strong support for hardware-in-the-loop simulations. Our robotic platforms, partially listed in Fig. 1, include G1 humanoids, Bunker robots with an edge-mounted arm, GEM Polaris autonomous vehicles, as well as various flying and ground robots.
Expected R&R of Korean SME Partner	As described above, we expect Korean SMEs to lead the selection of high-impact application scenarios for our theory and proposed technologies. We will then jointly pursue a full-scale demonstration of our approach. We also anticipate that SMEs will lead the production of the resulting prototype technologies, potentially advancing them to TRL 5.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Our project will provide a formal tool to prove—or disprove—the feasibility of the order-of-magnitude gains widely reported for vision- and language-infused dynamical systems across domains. By demonstrating it in high-impact applications selected by SMEs, we can further establish its broader potential across a wide range of domains. This will include vision-based proximity operations for high-precision drone delivery and infrastructure inspection, time-sensitive vision and language processing for environmental monitoring and disaster response, and safety-critical autonomous vehicle coordination in human-centered environments, just to name a few.

6. References

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